The Project Title

A Study on Speech Emotion Recognition Based on Multi-Feature and Deep Models

Matthew Leeke

School of Computer Science

College of Engineering and Physical Sciences

University of Birmingham

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Abstract

Acknowledgements

Abbreviations

ACB	Apple Banana Carrot

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CHAPTER 1

Introduction

1.0.1 Background

Speech conveys rich affective information through prosody, articulation, and spectral characteristics, enabling listeners to infer speakers' intentions and internal states. Speech Emotion Recognition (SER) aims to computationally infer human emotion from acoustic signals, either as discrete categories (e.g., anger, happiness, sadness, fear, and neutral) or along continuous dimensions (e.g., valence, arousal, and dominance). Rooted in speech processing, affective computing, and psychology, early SER systems relied on handcrafted prosodic and spectral features combined with classical classifiers. Over the past decade, deep learning has transformed the field: convolutional networks on time–frequency representations, recurrent and attention-based models for temporal dynamics, and, more recently, transformer architectures and self-supervised pretraining have markedly improved generalization. The research focus has also shifted from acted laboratory corpora to spontaneous, in-the-wild data, emphasizing robustness to noise, channel variability, and cultural–linguistic diversity. In parallel, there is growing attention to transparency, privacy, and fairness, reflecting the societal implications of emotion inference technologies.

SER has broad practical value. In customer service and contact centers, emotion-aware analytics can monitor user satisfaction and support agent coaching in real time. In healthcare and mental health, non-invasive acoustic markers assist in screening and longitudinal monitoring of affective disorders, stress, and burnout. In human–computer interaction, conversational assistants and social robots can adapt responses and dialogue strategies based on users' emotional cues. In automotive and smart environments, continuous monitoring of driver frustration or fatigue enhances safety and personalization. Education technology, entertainment, and recommendation systems also benefit from affect-aware feedback and content curation. These applications motivate solutions that are accurate, low-latency, and resource-efficient for both cloud and on-device deployment.

The research significance of SER is twofold. Scientifically, SER advances our understanding of how affect is encoded in speech and how to learn robust, interpretable representations under limited, imbalanced, and noisy supervision. Technically, it drives innovation in domain generalization, cross-lingual transfer, and multimodal fusion with text and vision, while promoting explainability to build user trust. Addressing ethical and legal considerations—including data privacy, informed consent, and bias mitigation—is essential to responsible deployment. Finally, rigorous benchmarking, reproducibility, and standardized evaluation protocols (e.g., unweighted/weighted accuracy for categorical tasks and concordance correlation for continuous estimation) are critical for cumulative progress. Altogether, SER research provides both theoretical insights and practical tools for empathetic, human-centered intelligent systems.

1.0.2 Main Contents of the Dissertation

This dissertation addresses the challenge of suboptimal recognition accuracy in Speech Emotion Recognition by developing a deep learning based framework that integrates speech signal preprocessing, feature engineering, attention mechanisms, and model optimization. We conduct a systematic study of SER methods from multiple perspectives, including audio preprocessing, data augmentation, feature selection, model architectures, and evaluation metrics. The central objective is to learn feature representations with stronger affective discriminability directly from speech, thereby improving accuracy, robustness, and generalization across speakers, recording conditions, and datasets. The proposed approach emphasizes practical deployability by balancing performance with computational efficiency for both cloud and on-device inference.

1.0.3 Structure of the Dissertation

The dissertation is organized as follows. Chapter 1 introduces the research background and motivation, identifies the key challenges in SER, and outlines the scope and structure of the thesis. Chapter 2 surveys related work on speech signal processing, feature extraction, the design of attention mechanisms, and emotion taxonomies and recognition paradigms, summarizing major international developments. Chapter 3 presents the methodology. We construct features from audio data using Mel-frequency cepstral coefficients (MFCCs) and Mel-spectrograms; we then investigate multiple models for SER, including a multilayer perceptron (MLP), a shallow convolutional neural network (CNN), ResNet-18 for 2D Mel-spectrogram inputs, and the Audio Spectrogram Transformer (AST), a ViT-based model for audio classification. Finally, we apply data augmentation by injecting noise during training to improve generalization and mitigate overfitting. Chapter 4 describes the experiments. We design comparative studies to evaluate the performance differences among features and models on SER tasks under consistent protocols and metrics. Chapter 5 provides discussion. We analyze the experimental results in depth, examine the relative strengths and weaknesses of different features and models, and articulate limitations and directions for future research. Chapter 6 concludes the thesis by summarizing the main findings and contributions, and highlighting potential avenues for subsequent work.

CHAPTER 2

Literature Review

2.1 Datasets for Speech Emotion Recognition

Datasets are foundational to empirical research in Speech Emotion Recognition (SER). They can be broadly categorized into acted corpora and spontaneous or in-the-wild corpora. Acted datasets are typically recorded in controlled laboratory environments with professional actors performing target emotions. While these corpora offer clean signals and clearer affective cues that facilitate modeling, they may over-represent exaggerated expressions and fail to capture the subtleties of everyday speech. By contrast, spontaneous corpora contain naturalistic affect produced in real interactions or semi-scripted dialogues. They better reflect practical deployment scenarios but introduce additional challenges such as background noise, speaker overlap, label ambiguity, and class imbalance. Below we summarize several widely used SER datasets.

RAVDESS. The Ryerson Audio-Visual Database of Emotional Speech and Song contains recordings from 24 professional actors (12 female, 12 male) producing two lexically matched statements across eight emotions (neutral, calm, happy, sad, angry, fearful, disgust, surprise) with normal and strong intensities (neutral has no intensity). The speech subset totals 1,440 audio files, recorded with consistent studio conditions, and is commonly used for speaker-independent evaluation. RAVDESS provides balanced classes and clear prosodic cues, making it suitable for benchmarking and ablation studies; however, its acted nature can limit ecological validity when models are deployed on spontaneous speech.

IEMOCAP. Collected at the University of Southern California, IEMOCAP is a multimodal corpus containing speech, video, facial motion capture, and transcripts over approximately 12 hours of dyadic interactions with 10 actors (five female, five male) arranged into five sessions (each session features one female–male pair). The corpus includes both scripted and improvised dialogues. Utterances are annotated by multiple raters for nine discrete emotions (anger, happiness, excitement, sadness, frustration, fear, surprise, neutral, and other) as well as three continuous dimensions (valence, arousal, dominance). In practice, many studies merge excitement into happiness, yielding a four-class set (neutral, angry, happy, sad) or a six-class set (neutral, angry, happy, sad, frustration, surprise). For comparability, this thesis adopts both the four-class and six-class protocols in line with prior work.

EmoDB. The Berlin Database of Emotional Speech comprises recordings from 10 German actors (five female, five male) who produced 10 phonetically balanced sentences with seven target emotions: anger, sadness, happiness, fear, neutral, disgust, and boredom. After rigorous quality screening, around 500 utterances were retained. Audio was recorded in a studio at 48 kHz, 16-bit, and is often downsampled to 16 kHz for processing. Although acted, the dataset emphasizes natural-sounding realizations through actor induction techniques, providing strong, well-separated emotional prototypes.

MELD. The Multimodal EmotionLines Dataset extends EmotionLines by adding audio and video modalities to textual dialogues. Derived from the TV show "Friends," MELD contains over 1,400 multi-speaker conversations and more than 13,000 utterances, annotated with seven emotions: anger, disgust, sadness, joy, neutral, surprise, and fear. Its multimodal, conversational structure enables the study of contextual emotion dynamics and speaker interactions beyond isolated utterances.

2.2 SER with Acoustic Features

Acoustic feature engineering plays a central role in early SER pipelines. It compresses raw waveforms into compact descriptors derived from time-domain and frequency-domain analyses, guided by speech science. Canonical categories include prosodic features (pitch, energy, duration), voice-quality features (jitter, shimmer, harmonicity), spectral features (formants, cepstral coefficients), and Teager Energy Operator (TEO) based features. These low-level descriptors (LLDs) capture frame-wise micro-variations in speech and can be summarized into high-level functional statistics (HSFs) such as means, first- and second-order derivatives, and other moments to encode utterance-level context.

Empirical studies have shown that acoustic features carry substantial affective information, reflecting changes in emotion, speaker identity, speaking style, and tempo [41]. As illustrated in Figure 1.2, LLDs and HSFs form the basis of many classical SER systems. Lee et al. and Schuller et al. reported strong results using feature sets combining energy-, pitch-, and spectral-related cues, with pitch-related features often proving more discriminative than energy features. Rao et al. [54] compared LLDs against HSFs (and their combination) and found that combining frame-level prosody (e.g., F0 trajectories, energy) with statistical summaries improves recognition performance. Lugger et al. integrated prosodic and voice-quality features (e.g., glottal gradient, skewness gradient, closure rate gradient) within a two-stage classifier to reach 74.5% accuracy. Li et al. further augmented HSFs with jitter and shimmer to boost performance. Zhang et al. combined prosodic with voice-quality features (including the first three formants), improving accuracy by about 10% over prosody-only baselines. Sato et al. used clustering to label frames and constructed a segmented MFCC feature set, outperforming KNN on prosody and HMMs on conventional MFCCs. Bitouk et al. introduced new spectral features by computing MFCCs separately for stressed vowels, unstressed vowels, and consonants, concluding that consonants may carry richer affective cues than vowels. Zhou et al. proposed three TEO variants—especially a standard-band TEO autocorrelation envelope—to study airflow energy changes under stress, achieving better results than pitch+MFCC combinations.

After feature extraction, classifiers are typically used for final affect prediction. Traditional approaches include Gaussian Mixture Models (GMMs), Support Vector Machines (SVMs), and Decision Trees (DTs) . More recent methods employ deep learning to learn discriminative affect representations from acoustic features, such as 1D CNNs and RNNs, trained with task-appropriate loss functions. Representative studies are summarized in Table 1.1.

Overall, acoustic-feature-based SER offers two key advantages. First, it provides a degree of interpretability: models can assign higher weights to features with stronger affective discriminability, which often aids accuracy compared to purely black-box baselines. Second, system complexity can be relatively low: feature extraction is algorithmic with few learnable parameters, while deep architectures built on these features improve adaptability and reduce overfitting, enabling transfer across related affective tasks. Despite progress, several issues remain: (1) many features rely on the short-time Fourier transform (STFT), a global transform that may inadequately represent the inherently non-stationary, localized time–frequency patterns of affect; (2) attention mechanisms operating on handcrafted features may struggle to highlight truly salient regions, potentially discarding relevant emotional cues.

2.3 SER with Spectrogram-based Features

A central question in spectrogram-based SER is how to extract affectively discriminative representations from time–frequency images. Mel-spectrograms differ from classical handcrafted features (e.g., formants, jitter, MFCCs) in that they preserve localized time–frequency structure and delegate hierarchical feature learning to deep networks. Representing energy over time and frequency in a 2D grid enables learning of complex spectro-temporal patterns that are difficult to encode with fixed handcrafted descriptors [63]. During training, networks can discover filter-like patterns akin to Mel filterbanks while also capturing long-range dependencies and contextual cues typically missed by shallow features.

Since the 2012 deep learning breakthroughs on ImageNet, advances in model capacity and data processing have translated to SER. Compared with conventional machine learning, deep models can learn affect representations without manual feature selection. Feed-forward deep neural networks (DNNs) comprise multiple hidden layers whose weights are optimized via gradient descent and backpropagation to minimize a task loss. Convolutional neural networks (CNNs) are particularly impactful in SER , as spectrograms exhibit strong local correlations along time and frequency; local receptive fields allow CNNs to extract progressively higher-level features from 2D inputs . Recurrent neural networks (RNNs) and Long Short-Term Memory (LSTM) networks are commonly stacked on top of CNN features to model temporal dependencies and can also operate directly on frame-level acoustic features to yield high-level affect representations . Attention mechanisms further refine these representations by emphasizing salient regions conducive to emotion discrimination.

Rapid progress has been reported in recent years . Trigeorgis et al. and Lim et al. proposed CNN–LSTM hybrids in which CNNs extract high-level spectrogram features and LSTMs capture sequence dynamics, outperforming traditional machine learning; similar architectures have been widely adopted. To model spatial relationships in spectra, capsule networks based on CNNs have also been explored , achieving promising sentence-level affect modeling. Representative spectrogram-based deep architectures for SER are summarized in Table 1.2.

On top of high-capacity encoders, attention mechanisms have been extensively used to further improve accuracy , including self-attention, local attention, hop-wise attention, and many other variants . These modules enhance weights of informative time–frequency regions while suppressing irrelevant ones, thereby focusing the network on discriminative patterns; see Table 1.3 for recent studies leveraging attention to boost SER.

Finally, learned affect representations are mapped to task outputs via suitable loss functions and evaluation metrics. For categorical SER, cross-entropy remains the dominant training objective, while mean squared error is often used for dimensional emotion regression. Historically, accuracy was the default metric; however, as datasets grew and class imbalance became prevalent, the Interspeech 2009 Emotion Challenge popularized unweighted average recall (UAR), computed as the mean of per-class recalls, which is now widely adopted for imbalanced settings. For dimensional SER, the concordance correlation coefficient (CCC) is commonly used. From Tables 1.3 and 1.4 we observe steady gains from stronger encoders and attention, yet several challenges remain: (1) fixed-size convolutional kernels can restrict the effective receptive field over salient emotional regions, limiting global context capture; (2) attention layers may over-rely on fully connected heads and softmax, under-exploiting local, global, and positional cues; (3) cross-entropy is sensitive to label distributions and can struggle when class manifolds are close; (4) variable-length utterances complicate segment-level training and can misalign segment labels with utterance-level annotations.

CHAPTER 3

Methodology

3.1 Feature Extraction

This section details the feature extraction pipeline and unified notation used throughout the thesis. Let $x[n]$ denote the discrete-time speech signal sampled at rate f_s (Hz). Frames are indexed by m, frequency bins by k, and Mel bands by f. The frame length is N samples, hop size is H samples, the analysis window is $w[n]$ (Hann unless otherwise stated), the discrete Fourier transform (DFT) length is K, and the number of Mel filters is F. We use $X[k, m]$ for the short-time spectrum, $P[k, m] = |X[k, m]|^2$ for the power spectrum, $\mathbf{B} \in \mathbb{R}^{F \times K}$ for the Mel filterbank, and a small constant $\varepsilon > 0$ for numerical stability.

3.1.1 MFCC

Mel-frequency cepstral coefficients (MFCCs) are compact features that approximate human auditory perception by (i) warping frequency to the Mel scale, (ii) applying a band-pass filterbank to obtain perceptual energies, (iii) compressing the dynamic range, and (iv) decorrelating energies via a discrete cosine transform (DCT). The standard pipeline is as follows.

1) Pre-emphasis (optional) enhances high-frequency components:

$$y[n] = x[n] - \alpha\, x[n-1], \quad \alpha \in [0.95, 0.99]. \tag{3.1}$$

2) Framing and windowing create approximately stationary short-time segments:

$$X[k, m] = \sum_{n=0}^{N-1} x[n + mH]\, w[n]\, e^{-j2\pi kn/K}, \quad 0 \le k < K. \tag{3.2}$$

3) Power spectrum is computed as

$$P[k, m] = \frac{|X[k, m]|^2}{K}. \tag{3.3}$$

4) Mel filterbank energies are obtained by triangular filters spaced on the Mel scale:

$$S[f,m] = \sum_{k=0}^{K-1} B[f,k]\, P[k,m], \quad 1 \le f \le F. \tag{3.4}$$

The Mel frequency mapping for a linear frequency ν (in Hz) is

$$\mathrm{mel}(\nu) = 2595\, \log_{10}\left(1 + \frac{\nu}{700}\right), \tag{3.5}$$

and the filterbank $\mathbf{B}$ is formed by piecewise-linear triangular filters uniformly spaced on the Mel axis and then mapped back to linear frequency bins.

5) Logarithmic compression reduces dynamic range and approximates loudness perception:

$$\widehat{S}[f,m] = \log\big(S[f,m] + \varepsilon\big). \tag{3.6}$$

6) DCT-II decorrelates the log-Mel energies, yielding MFCCs:

$$c_q[m] = \sum_{f=1}^{F} \widehat{S}[f,m] \cos\!\left(\frac{\pi q}{F}\left(f - \tfrac{1}{2}\right)\right), \quad 0 \le q < Q, \tag{3.7}$$

where Q is the number of cepstral coefficients retained (commonly $Q \in [12, 20]$). An optional sinusoidal lifter improves energy distribution across coefficients:

$$\widetilde{c}_q[m] = \left(1 + \frac{L}{2} \sin\frac{\pi q}{L}\right) c_q[m], \quad L > 0. \tag{3.8}$$

Temporal dynamics are often captured by regression-based derivatives using a symmetric window of radius R:

$$\Delta c_q[m] = \frac{\sum_{r=1}^{R} r\left(c_q[m+r] - c_q[m-r]\right)}{2\sum_{r=1}^{R} r^2}, \quad \Delta\Delta c_q[m] \text{ analogously.} \tag{3.9}$$

Advantages: MFCCs are compact, relatively robust to slowly varying convolutive effects, and widely validated in speech tasks. Limitations: information loss due to coarse spectral integration and DCT decorrelation, frame-level stationarity assumptions, and sensitivity to noise and channel mismatch without appropriate normalization or enhancement.

3.1.2 Mel-spectrogram

The Mel-spectrogram represents short-time spectral energy projected onto a perceptual frequency axis, typically used directly as a 2D input to convolutional or transformer models.

Given the short-time spectrum in (3.2) and power spectrum in (3.3), the Mel-spectrogram is

$$\mathrm{MelSpec}[f,m] = \sum_{k=0}^{K-1} B[f,k]\, P[k,m], \quad 1 \le f \le F. \tag{3.10}$$

A log or power-law compression is commonly applied to stabilize training and approximate loudness:

$$\mathrm{LogMel}[f,m] = \log\big(\mathrm{MelSpec}[f,m] + \varepsilon\big) \quad \text{or} \quad \mathrm{MelSpec}[f,m]^{\gamma},\ \gamma \in (0,1]. \tag{3.11}$$

Design choices include the sample rate f_s, frame length N, hop size H, window type $w[n]$, number of Mel filters F, lower/upper frequency bounds of the filterbank, and whether to use magnitude or power spectra. Per-utterance or global mean-variance normalization is often applied to reduce channel variability.

Advantages: Mel-spectrograms preserve local time–frequency structure and are highly compatible with CNNs and attention-based encoders; they avoid the information loss introduced by the DCT in MFCC. Limitations: higher dimensionality increases memory and compute cost; the representation remains sensitive to additive noise and reverberation without enhancement or robust training; choices of F, bounds, and f_s affect resolution and cross-dataset transferability.

3.2 Model Developments

We consider a label set $\mathcal{C}$ with $|\mathcal{C}| = C$ emotion classes. For utterance-level classification, we denote the per-frame MFCC feature as $\mathbf{z}_m \in \mathbb{R}^Q$ from (3.7) (optionally concatenated with Δ and $\Delta\Delta$ features from (3.9)). A fixed-dimensional utterance representation is obtained by temporal mean pooling

$$\bar{\mathbf{z}} = \frac{1}{T}\sum_{m=1}^{T} \mathbf{z}_m \in \mathbb{R}^D, \quad (D = Q \text{ or } 3Q), \tag{3.12}$$

optionally augmented with standard deviation pooling. Let $\mathbf{y} \in \{0,1\}^C$ be the one-hot label vector.

3.2.1 Multi-layer Perceptron

We adopt a feed-forward multilayer perceptron (MLP) as a lightweight baseline operating on MFCC-based utterance vectors. Given input $\bar{\mathbf{z}}$ from (3.12), a depth-L MLP computes

$$\mathbf{h}^{(1)} = \sigma\big(\mathbf{W}^{(1)}\bar{\mathbf{z}} + \mathbf{b}^{(1)}\big), \tag{3.13}$$

$$\mathbf{h}^{(\ell)} = \sigma\big(\mathbf{W}^{(\ell)}\mathbf{h}^{(\ell-1)} + \mathbf{b}^{(\ell)}\big), \quad 2 \le \ell \le L-1, \tag{3.14}$$

$$\mathbf{o} = \mathbf{W}^{(L)}\mathbf{h}^{(L-1)} + \mathbf{b}^{(L)}, \tag{3.15}$$

where $\sigma(\cdot)$ is a pointwise nonlinearity (ReLU by default), $\mathbf{W}^{(\ell)}$ and $\mathbf{b}^{(\ell)}$ are trainable parameters. Class probabilities are

$$\mathbf{p} = \mathrm{softmax}(\mathbf{o}), \quad p_c = \frac{\exp(o_c)}{\sum_{c'=1}^{C}\exp(o_{c'})}. \tag{3.16}$$

We train with cross-entropy and ℓ_2 regularization:

$$\mathcal{L} = -\sum_{c=1}^{C} y_c \log p_c + \lambda\|\Theta\|_2^2, \tag{3.17}$$

where Θ collects all parameters and λ controls weight decay. Dropout between hidden layers improves generalization.

Role in this thesis: the MLP serves as a parameter-efficient baseline with MFCC inputs, quantifying the incremental benefits of time–frequency 2D models.

3.2.2 Shallow CNN

For time–frequency inputs we use the log-Mel representation $\mathrm{LogMel} \in \mathbb{R}^{F\times T}$ from (3.11). A shallow 2D CNN treats LogMel as a single-channel image: $\mathbf{X} \in \mathbb{R}^{1\times F\times T}$. Let a 2D convolution with kernel $\mathbf{W} \in \mathbb{R}^{C_{\text{out}}\times C_{\text{in}}\times k_f\times k_t}$, stride (s_f, s_t), and padding (p_f, p_t) be

$$(\mathbf{W} * \mathbf{X})[c, f, t] = \sum_{c'=1}^{C_{\text{in}}} \sum_{i=0}^{k_f-1} \sum_{j=0}^{k_t-1} \mathbf{W}[c, c', i, j]\, \mathbf{X}[c', f+i-p_f,\, t+j-p_t]. \tag{3.18}$$

Each convolutional block applies convolution, batch normalization, ReLU, and pooling:

$$\mathbf{U}_1 = \mathrm{ReLU}\big(\mathrm{BN}(\mathbf{W}_1 * \mathbf{X})\big), \qquad \mathbf{V}_1 = \mathrm{Pool}(\mathbf{U}_1), \tag{3.19}$$
$$\mathbf{U}_2 = \mathrm{ReLU}\big(\mathrm{BN}(\mathbf{W}_2 * \mathbf{V}_1)\big), \qquad \mathbf{V}_2 = \mathrm{Pool}(\mathbf{U}_2). \tag{3.20}$$

We then apply global average pooling over time and frequency and a linear classifier to obtain logits $\mathbf{o}$ followed by (3.16).

Design: we use two convolutional blocks with moderate kernels (e.g., $k_f \in \{3, 5\}$, $k_t \in \{3, 5\}$), stride 1, and max-pooling with small windows to preserve resolution. Regularization includes dropout and additive noise during training to improve robustness.

Advantages: shallow CNNs are data-efficient and fast, capturing local spectro-temporal patterns. Limitations: weaker long-range modeling and limited receptive field without deeper stacks or dilations. In this thesis, the shallow CNN consumes log-Mel features to quantify the benefits of local 2D structure over MFCC-based MLPs.

3.2.3 ResNet-18

We adopt ResNet-18 as a deeper convolutional baseline to enhance receptive field and hierarchical feature learning on LogMel. A basic residual block computes

$$\mathbf{y} = \mathcal{F}(\mathbf{x};\, \{\mathbf{W}_i\}) + \mathcal{S}(\mathbf{x}), \tag{3.21}$$

where $\mathcal{F}$ is a stack of Conv→BN→ReLU→Conv→BN, and $\mathcal{S}$ is either identity or a 1×1 projection to match shape when stride/downsampling is used. The network stacks such blocks with increasing channels and occasional stride-2 to downsample along time and frequency. After global average pooling, a linear classifier produces logits $\mathbf{o}$ and probabilities via (3.16).

For spectrogram inputs, the first convolution is adapted to single-channel input ($C_{\text{in}} = 1$) with kernel size 7×7 and stride 2, followed by batch normalization and ReLU. Weight decay and label smoothing may be applied to improve generalization.

Advantages: ResNet-18 balances capacity and efficiency, modeling broader context and invariances. Limitations: increased compute relative to shallow CNNs and potential overfitting on small corpora without strong regularization. In this thesis, ResNet-18 processes LogMel to assess the gains of residual learning.

3.2.4 Audio Spectrogram Transformer

The Audio Spectrogram Transformer (AST) adapts the Vision Transformer (ViT) to audio by treating the log-Mel spectrogram as a 2D patch sequence. Given LogMel $\in \mathbb{R}^{F\times T}$, we partition it into non-overlapping patches of size $P_f \times P_t$, yielding $N = \lfloor F/P_f \rfloor \cdot \lfloor T/P_t \rfloor$ patches. Each patch is flattened and linearly projected to a token embedding:

$$\mathbf{z}_i = \mathrm{vec}(\mathrm{patch}_i)\,\mathbf{W}_e + \mathbf{b}_e \in \mathbb{R}^d, \quad i = 1, \ldots, N. \tag{3.22}$$

A learnable classification token $\mathbf{z}_{\mathrm{cls}}$ is prepended and positional embeddings $\mathbf{p}_i$ are added. The resulting sequence $\{\mathbf{z}_{\mathrm{cls}}, \mathbf{z}_1, \ldots, \mathbf{z}_N\}$ is processed by a stack of transformer encoder layers, each composed of multi-head self-attention (MHSA) and a positionwise feed-forward network (FFN) with residual connections and layer normalization. For a single head,

$$\mathrm{Attn}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \mathrm{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d_k}}\right)\mathbf{V}, \tag{3.23}$$

with queries $\mathbf{Q} = \mathbf{Z}\mathbf{W}^Q$, keys $\mathbf{K} = \mathbf{Z}\mathbf{W}^K$, values $\mathbf{V} = \mathbf{Z}\mathbf{W}^V$, and $\mathbf{Z}$ the input token matrix; MHSA concatenates multiple heads. The FFN uses two linear layers and a nonlinearity (e.g., GELU). The classification head reads the [CLS] token to produce logits $\mathbf{o}$ and probabilities via (3.16).

Design: patch sizes (P_f, P_t) balance spectral and temporal resolution; smaller patches increase sequence length and cost but improve detail. Regularization includes dropout and stochastic depth; data augmentation includes time/frequency masking and additive noise.

Advantages: AST captures global dependencies and long-range context beyond CNN receptive fields. Limitations: higher computational cost and sensitivity to data scale; careful regularization and pretraining can be beneficial. In this thesis, AST consumes log-Mel inputs to examine the benefits of attention-based modeling.

3.3 Data Augmentation

Deep models for SER often overfit clean, laboratory speech and degrade under domain shift caused by background noise, channel mismatch, and reverberation. To improve generalization from clean training conditions to noisy evaluations, we perform on-the-fly additive-noise augmentation during training while keeping the validation and clean test sets unchanged. We follow the unified notation in Section 6: the discrete-time speech signal is $x[n]$ at sample rate f_s, and we denote a noise signal by $v[n]$. All mixing is performed in the time domain prior to feature extraction (MFCC or Log-Mel), ensuring that downstream features consistently reflect the augmented acoustics.

3.3.1 Noise sources and SNR set

We consider two noise families: (i) white Gaussian noise as a controlled baseline; and (ii) environmental recordings sampled from ESC-50 (three representative classes). The target signal-to-noise ratio (SNR) in decibels is uniformly sampled from $\mathcal{S} = \{0, 5, 10, 20\}$ dB. When comparing architectures, we additionally use a fixed 20 dB white-noise setting as a reference point.

3.3.2 On-the-fly sampling policy

For each training utterance, with probability $p_{\text{aug}} = 0.7$ we inject one type of noise, otherwise we keep the sample clean. The noise family (white vs. ESC-50 subset) and the SNR are sampled uniformly. Each speech sample is mixed with at most one noise instance.

3.3.3 Time alignment and preprocessing

Given a speech clip of length T samples and a noise recording $v[n]$: if the noise is shorter than T, we tile it by concatenation or loop with a random starting offset; if it is longer, we randomly crop a segment of length T. Both speech and noise are resampled to f_s if needed and centered to zero mean. Let $x_s[n]$ and $x_n[n]$ denote the length-matched speech and noise signals, respectively, for $0 \leq n < T$.

3.3.4 RMS, target SNR, and gain computation

We use the root-mean-square (RMS) to control the noise gain. For a discrete signal $z[n]$ of length T:

$$R(z) = \sqrt{\frac{1}{T} \sum_{n=0}^{T-1} z[n]^2}. \tag{3.24}$$

Let $R_s = R(x_s)$ and $R_n = R(x_n)$. Given a desired SNR in decibels, SNR_{dB}, the linear SNR is $\text{SNR}_{\text{lin}} = 10^{\text{SNR}_{\text{dB}}/20}$. We choose a scalar gain g such that the mixture achieves the target SNR:

$$g = \frac{R_s}{R_n \, \text{SNR}_{\text{lin}}} = \frac{R_s}{R_n \cdot 10^{\text{SNR}_{\text{dB}}/20}}. \tag{3.25}$$

The mixture is

$$x_{\text{mix}}[n] = x_s[n] + g \, x_n[n]. \tag{3.26}$$

By construction, the realized SNR satisfies

$$\text{SNR}_{\text{dB}} = 20 \, \log_{10}\left(\frac{R(x_s)}{R(g \, x_n)} \right) = 20 \, \log_{10}\left(\frac{R_s}{g \, R_n} \right). \tag{3.27}$$

3.3.5 Peak limiting and re-normalization

To prevent clipping after mixing, we enforce a peak limit of -1 dBFS. Let $A_{\max} = 10^{-1/20}$ be the amplitude threshold and $A_{\text{peak}} = \max_n |x_{\text{mix}}[n]|$. If $A_{\text{peak}} > A_{\max}$, we rescale

$$\widetilde{x}_{\text{mix}}[n] = x_{\text{mix}}[n] \, \frac{A_{\max}}{A_{\text{peak}}}, \tag{3.28}$$

and otherwise leave the mixture unchanged. A final mean-variance normalization per utterance can be applied before feature extraction for numerical stability.

3.3.6 Integration with the training pipeline

The augmentation is performed online per mini-batch before computing features: we form either MFCCs or Log-Mel spectrograms from $\widetilde{x}_{\text{mix}}[n]$ or $x_s[n]$ depending on whether augmentation was applied. This preserves consistency of notation and ensures identical downstream preprocessing. The same policy is shared by all architectures (MLP, shallow CNN, ResNet-18, AST) to enable fair comparison.

3.3.7 Evaluation protocol

We adopt RAVDESS (speech actors 01–24) as the primary dataset with the existing data split strategy. Augmentation is applied only to the training split. We evaluate on: (i) the original clean test set to verify no degradation on matched conditions; and (ii) noisy test sets generated by mixing the clean test audio with held-out noise at SNRs in $\mathcal{S}$. We report standard metrics for categorical SER (e.g., accuracy and unweighted average recall) and analyze performance as a function of SNR to identify robustness trends and breakpoints. A fixed 20 dB white-noise protocol is additionally used as a reference to compare architectural robustness, with particular attention to transformer-based AST versus conventional CNN/MLP baselines.

CHAPTER 4

Experiments

This chapter presents a comprehensive experimental study on Speech Emotion Recognition (SER) under clean and noisy conditions. We evaluate multiple architectures and noise augmentation strategies on two datasets, adhering to the unified notation and preprocessing defined in Chapter 3. Unless otherwise specified, audio is sampled at f_s, the time-domain signal is $x[n]$, noise is $v[n]$, and on-the-fly mixtures during training follow $x_{\text{mix}}[n] = x_s[n] + g\,x_n[n]$ with gain g computed from target SNR as in Chapter 3.

4.1 Common Settings

4.1.1 Datasets and splits

We use RAVDESS (speech actors 01–24) and IEMOCAP. Data splits follow the established protocol for each dataset. Validation and clean test sets are never augmented; only the training split is subjected to on-the-fly noise mixing when enabled.

4.1.2 Models

We compare a convolutional baseline (ResNet-18 on Log-Mel spectrograms) and a transformer-based model (AST Base on filterbank features), alongside lighter baselines where applicable (MLP and shallow CNN) to contextualize complexity–performance trade-offs.

4.1.3 Training and evaluation

Training uses mini-batch stochastic optimization with standard regularization. Metrics on the validation set include accuracy, precision, recall, and macro-F1. Confusion matrices are reported to analyze class-wise separability. Robustness is further assessed on noisy test sets created by offline mixing of $x[n]$ with held-out noise at specified SNRs $\in \{0, 5, 10, 20\}$ dB. All experiments fix a base random seed unless otherwise stated; key findings are validated by repeating with multiple seeds to estimate mean and variance.

4.1.4 Noise augmentation policy

When enabled, augmentation follows Chapter 3: with probability p_{aug}, we sample a noise type (white or ESC-50 subsets), sample SNR from $\{0, 5, 10, 20\}$ dB, time-align noise to the utterance, compute gain g via RMS matching to satisfy the target SNR, form $x_{\text{mix}}[n]$, apply peak limiting at -1 dBFS, and optionally mean–variance normalize prior to feature extraction.

4.2 Experiment A: Clean Baselines

Objective. Establish performance ceilings under matched clean conditions and provide a reference for subsequent noisy training.

Setup. No augmentation: all training, validation, and test audio remain clean. Features follow Chapter 3 (MFCC/Log-Mel) per model. Optimization settings are dataset-appropriate and fixed across runs.

Evaluation. Report accuracy, precision, recall, macro-F1 on validation and clean test sets. Provide confusion matrices to examine class confusability. Analyze overfitting by comparing training vs. validation curves.

Expected analysis. Identify baseline strengths and failure modes (e.g., emotions with overlapping prosody). Quantify headroom for robustness improvements.

4.3 Experiment B: White-Noise Training at 20 dB

Objective. Assess whether training with mild additive white noise (SNR = 20 dB) improves robustness without harming clean-set performance.

Setup. During training, with probability p_{aug} we mix white noise at SNR = 20 dB using gain from RMS control as defined in Chapter 3. Validation and test remain clean; an additional noisy test set at 20 dB is created offline for robustness assessment.

Evaluation. Compare metrics to Experiment A on clean validation/test, and on the 20 dB noisy test. Include confusion matrices to visualize changes in class boundaries.

Expected analysis. Determine whether small stochastic perturbations act as regularization that improves generalization. Inspect any trade-off on clean performance.

4.4 Experiment C: ESC-50 Multi-Category, Multi-SNR

Objective. Measure the effect of realistic environmental noise on SER across a range of SNRs and categories (natural soundscapes and human non-speech).

Setup. Training-time augmentation samples a noise category uniformly from two ESC-50 groups and SNR uniformly from $\{0, 5, 10, 20\}$ dB. Mixing follows the RMS-controlled procedure to obtain $x_{\text{mix}}[n]$ with peak limiting at -1 dBFS.

Evaluation. On the clean validation/test sets (unaugmented) and on noisy test sets stratified by noise category and SNR. Report per-SNR metrics and class-wise performance; provide heatmaps summarizing model $\times$ noise $\times$ SNR.

Expected analysis. Identify which noise categories and SNR regimes most strongly impact recognition. Examine whether models learn noise-invariant features.

4.5 Experiment D: SNR Ablation (Natural Soundscapes)

Objective. Characterize performance sensitivity to SNR under a fixed noise category to locate robustness breakpoints.

Setup. Train four models with the same natural-soundscape noise but a single SNR each: $0, 5, 10, 20$ dB. All other settings are identical.

Evaluation. Plot accuracy and macro-F1 versus SNR; compare confusion matrices across SNRs to see which emotions become confusable at low SNR.

Expected analysis. Determine the SNR threshold below which performance collapses, and quantify any non-linear degradation.

4.6 Experiment E: Model Robustness Comparison (ResNet-18 vs AST)

Objective. Compare architectural robustness between a CNN (ResNet-18) and a transformer (AST) under identical augmentation policies.

Setup. Use the multi-category, multi-SNR ESC-50 augmentation during training for both models. Features follow their respective best practices (Log-Mel for ResNet-18; filterbank features for AST). Keep validation clean; prepare noisy test sets matching the augmentation distribution for controlled comparison.

Evaluation. Report clean vs. noisy metrics side-by-side; include confusion matrices to highlight shifts in decision boundaries.

Expected analysis. Assess whether AST exhibits superior noise robustness due to global receptive fields and attention, and whether gains persist across datasets.

4.7 Experiment F: Augmentation Intensity Ablation (p_{aug})

Objective. Find the optimal augmentation probability by varying $p_{\text{aug}} \in \{0.3, 0.5, 0.7, 0.9\}$ under the multi-SNR ESC-50 setting.

Setup. Keep all factors fixed except p_{aug}. Training follows the same on-the-fly RMS-controlled mixing and peak limiting.

Evaluation. Plot performance versus p_{aug} and analyze under- vs over-augmentation effects. Inspect learning curves for signs of regularization benefits or optimization instability.

Expected analysis. Identify the augmentation intensity that maximizes robustness without harming clean accuracy; quantify variance across seeds.

4.8 Summary of Results and Visualization

We summarize results across models, noise types, SNRs, and datasets. Visualizations include: (i) performance–SNR curves per model; (ii) bar charts comparing macro-F1 for clean vs. noisy protocols; and (iii) confusion matrices for key settings to examine class-specific trends. Where applicable, we report mean $\pm$ standard deviation across seeds to reflect statistical stability.

CHAPTER 5

Discussion

CHAPTER 6

Conclusion

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